BAO CAO DO AN

Bai Kiem Tra So 3: Hoc Khong Giam Sat

Ung Dung Thuat Toan Unsupervised Learning Phan Cum Du Lieu NYC Airbnb

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Bo Du Lieu: New York City Airbnb Open Data

Nguon: Kaggle / Inside Airbnb | 20,758 listings | 28 features

Thuat toan: K-Means Clustering | DBSCAN Clustering

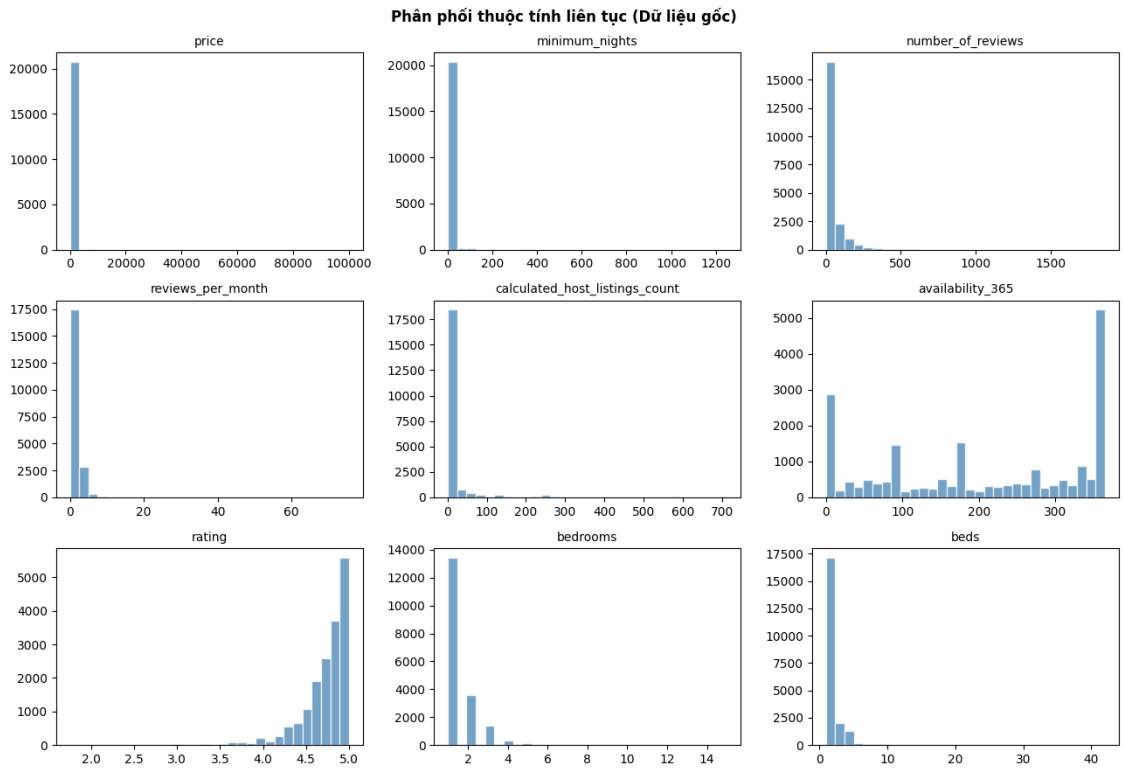
1. Mo Ta Du Lieu va Tien Xu Ly

1.1 Gioi Thieu Bo Du Lieu

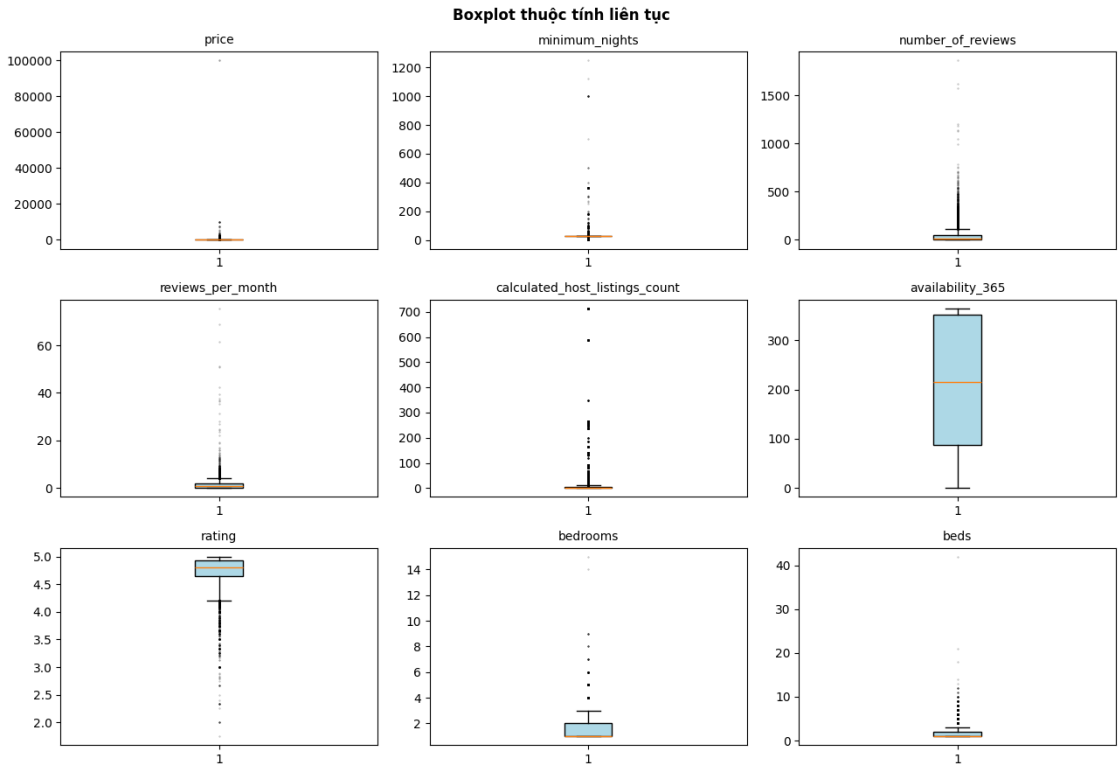
Bo du lieu su dung la **New York City Airbnb Open Data** - bo du lieu cong khai chua thong tin cac listing cho thue ngan han tai New York City. Day la bo du lieu dien hinh cho cac bai toan phan cum vi co su da dang cao ve kieu du lieu va phan phoi.

Thuoc tinh	Gia tri
So mau (samples)	20,758 listings
So thuoc tinh goc (preprocess)	22 cot
So thuoc tinh sau xu ly (postprocess)	28 features
Cot dinh danh	id, name, host_id, host_name
Cot lien tuc	price, latitude, longitude, rating, bedrooms, beds, baths, ...
Cot phan loai (da encode)	room_type (3 loai), neighbourhood_group (5 quan)
Missing values	Da xu ly hoan toan (0 sau xu ly)

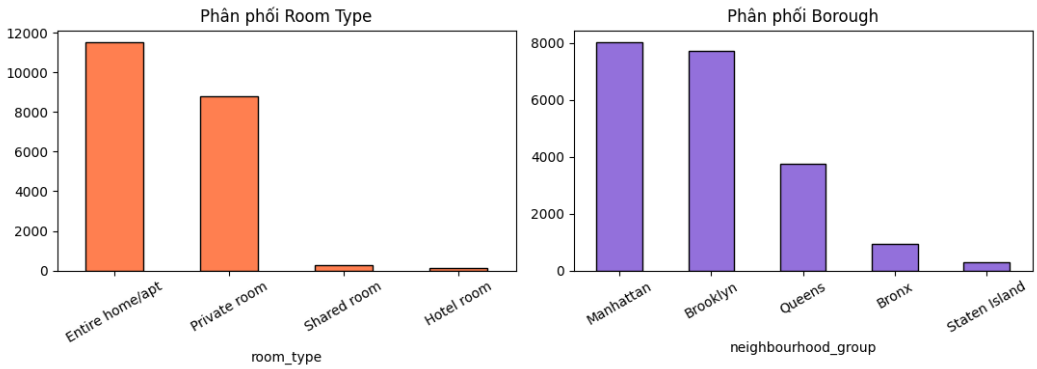
1.2 Phan Phoi Du Lieu



Hinh 1: Phan phoi cac thuoc tinh lien tuc cua du lieu goc



Hinh 2: Boxplot the hien outlier cua cac thuoc tinh



Hinh 3: Phan phoi bien phan loai - Room Type va Borough

2. Quy Trinh Tien Xu Ly Du Lieu

2.1 Cac Buoc Xu Ly

Du lieu sau khi thu thap tu nguon Airbnb duoc qua cac buoc tien xu ly sau:

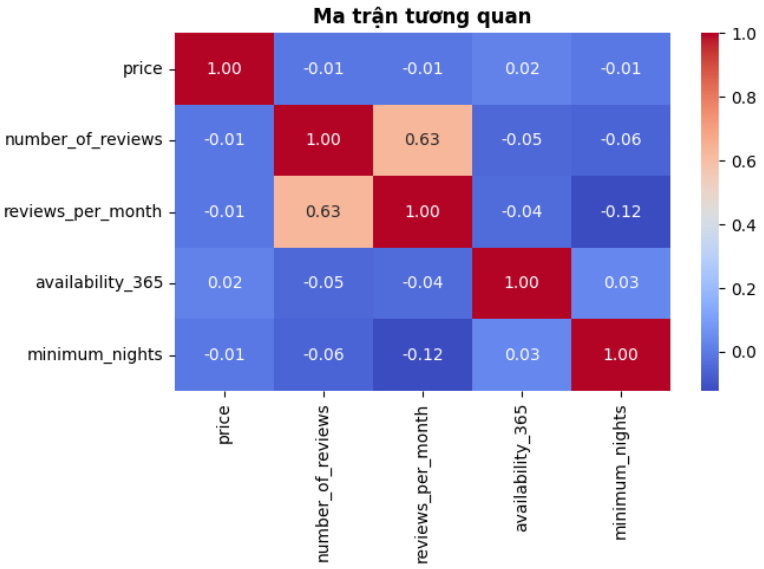
(1) **Xu ly gia tri thieu (Missing Values)**: Cac cot nhu 'last_review', 'reviews_per_month' co missing values - duoc xu ly bang cach dien 0 hoac loai bo dong khong hop le. Sau xu ly, postprocess_data.csv co 0 missing values.

(2) **Chuyen ma bien phan loai (Encoding)**: room_type va neighbourhood_group duoc chuyen sang One-Hot Encoding tao them 7 cot nhi phan. Cot 'has_license' duoc ma hoa nhi phan (0/1).

(3) **Feature Engineering**: Tao them price_log va calculated_host_listings_count_log de giam do lech cua phan phoi long-tail. Danh dau is_outlier cho 5 thuoc tinh chinh.

(4) **Chuan hoa (Scaling)**: Ap dung StandardScaler cho cac thuoc tinh lien tuc de dua ve cung thang do, tranh thuoc tinh co gia tri lon anh huong qua nhieu den ket qua.

2.2 Ma Tran Tuong Quan



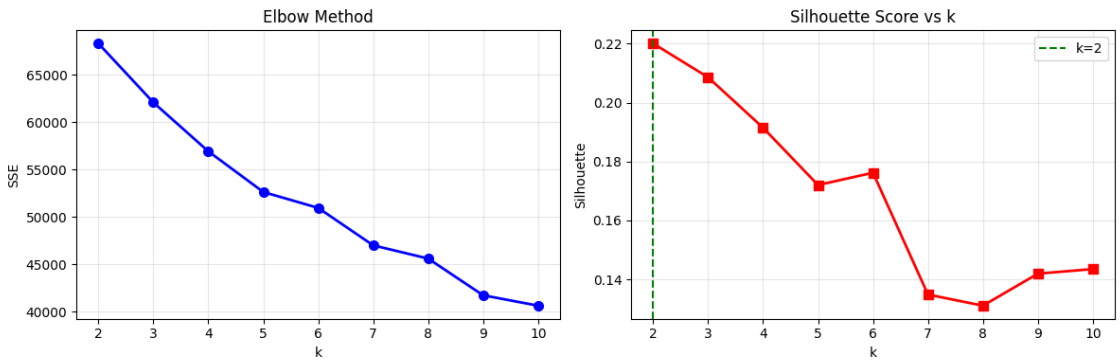
Hinh 4: Ma tran tuong quan cac thuoc tinh chinh

Nhan xet: reviews_per_month va number_of_reviews co tuong quan kha cao (duong tinh). Price co tuong quan am voi number_of_reviews - cac listing gia cao thuong co it luot danh gia hon.

3. Phan Cum voi K-Means

3.1 Chon So Cum K Toi Uu

Ap dung **Elbow Method** (SSE - Sum of Squared Errors) va **Silhouette Score** voi k tu 2 den 10 tren mau 5,000 diem:

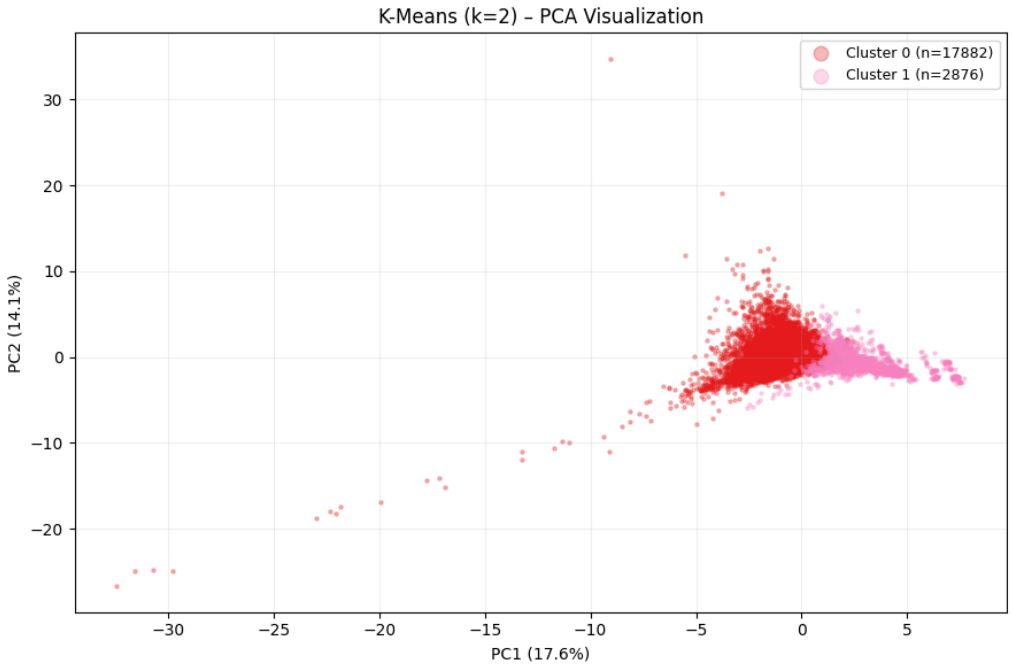


Hinh 5: Elbow Method va Silhouette Score theo k. K toi uu = 2

Ket luan: Silhouette Score dat gia tri cao nhat tai k = **2** (Score = 0.2243). Duong Elbow cung cho thay diem 'gap' ro rang tai day. Do do chon **k = 2** la so cum toi uu.

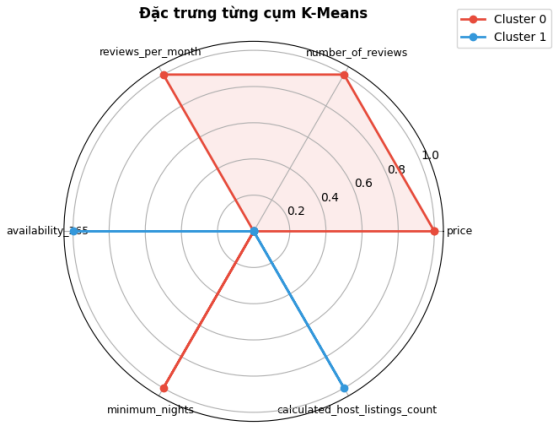
3.2 Ket Qua Phan Cum K-Means

Cluster	So luong mau	Ti le (%)
Cluster 0	17882	86.1%
Cluster 1	2876	13.9%

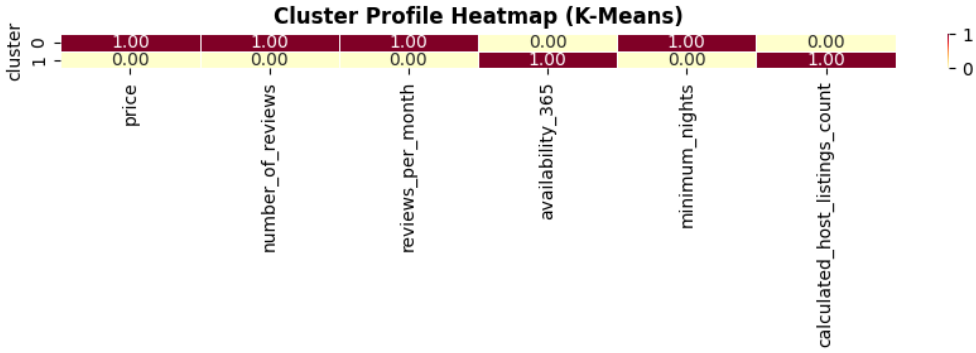


Hinh 6: Truc quan K-Means clusters tren khong gian PCA 2D

3.3 Dac Trung Tung Cum (K-Means)



Hinh 7: Radar Chart the hien profile cua tung cluster



Hinh 8: Heatmap gia tri trung binh cua tung cluster (chuan hoa 0-1)

Thuoc tinh	Cluster 0	Cluster 1
price	193.7	150.9
number_of_reviews	47.9	10.1
reviews_per_month	1.4	0.5
availability_365	200.8	238.4
minimum_nights	29.0	26.1

Nhan xet Cluster Profile:

Cluster 0 - Nhom Listings Thuong: Bao gom 17,882 listings (86.1%). Co gia trung binh ~$194/dem, so luot reviews nhieu (~48 reviews), tan suat reviews cao (1.38/thang), thoi gian available trung binh (~201 ngay/nam). Day la nhom listing ca nhan voi 1 host 1-2 properties, thuong la Entire home hoac Private room o Brooklyn/Manhattan.

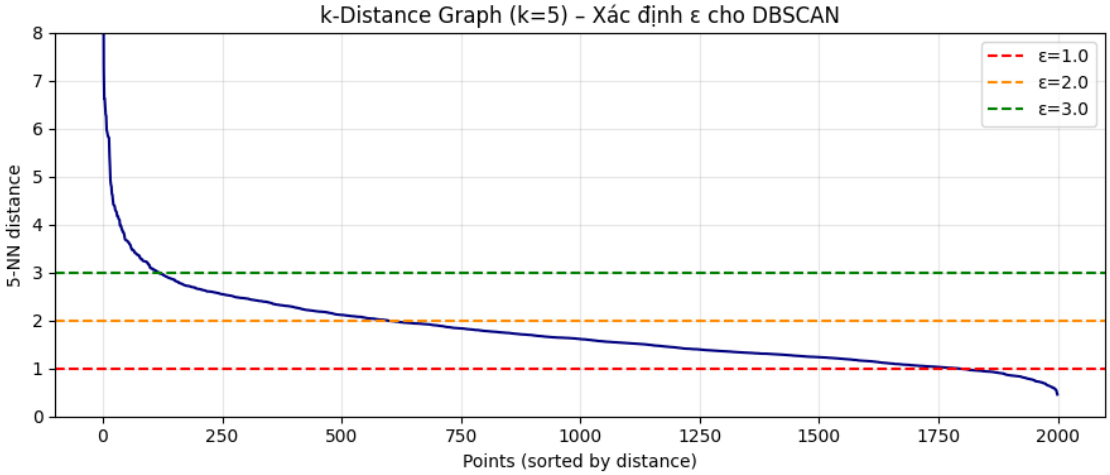
Cluster 1 - Nhom Host Chuyen Nghiep (Professional Hosts): 2,876 listings (13.9%). Co gia thap hon (~$151/dem) nhung so luong listings rat cao (trung binh 119 listings/host). So luot

review it (~10 reviews), co the la cac property moi hoac it pho bien. Day la nhom cac Professional Hosts / Property Managers voi nhieu listings.

4. Phan Cum voi DBSCAN

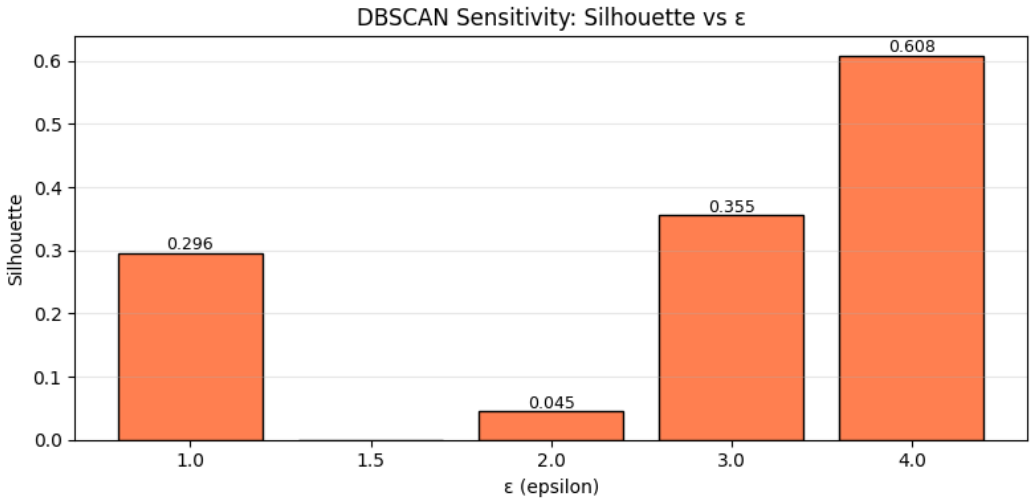
4.1 Chon Tham So Epsilon va min_samples

Su dung k-Distance Graph (k=5) de xac dinh nguong epsilon phu hop. Phuong phap nay sap xep cac diem theo khoang cach toi lang gieng thu 5 (giam dan) va tim diem 'gap' (elbow) - day chinh la gia tri epsilon can chon.



Hinh 9: k-Distance Graph (k=5). Cac duong ke thi nghiem voi eps = 1.0, 2.0, 3.0

4.2 Phan Tich Do Nhay Tham So DBSCAN



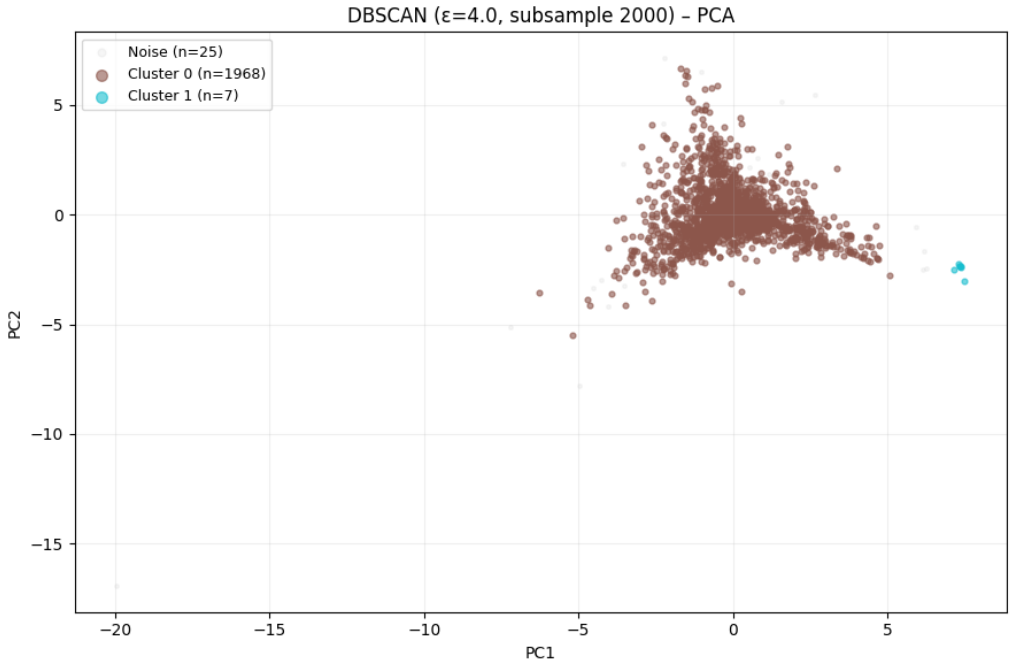
Hinh 10: Silhouette Score khi thay doi epsilon (min_samples=5)

epsilon	So cum	Noise points	Noise (%)	Silhouette
1.0	24	1641	82.0%	0.2956
1.5	13	911	45.6%	N/A
2.0	8	407	20.3%	0.0453
3.0	3	74	3.7%	0.3550

4.0	2	25	1.2%	0.6082

Tham so da chon: epsilon = 4.0, min_samples = 5. Ket qua: 2 clusters, 25 noise points (1.2% du lieu mau). Silhouette Score = 0.6082.

4.3 Ket Qua Truc Quan DBSCAN



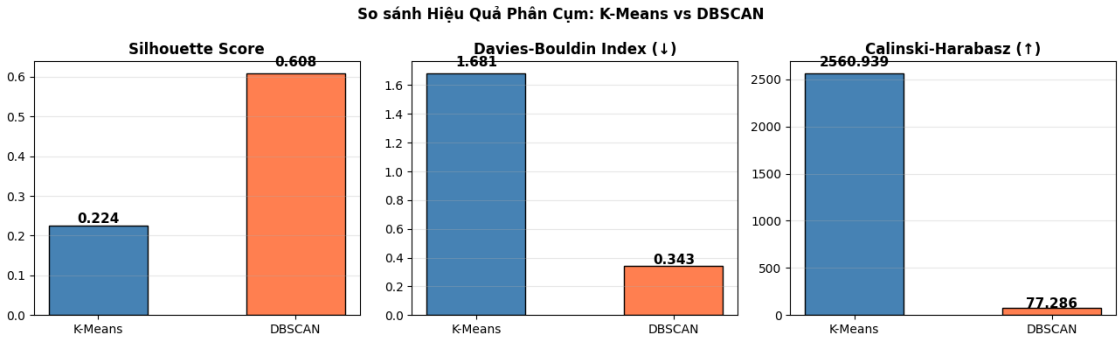
Hinh 11: DBSCAN (eps=4.0) tren PCA 2D. Diem xam = noise/outlier

Nhan xet DBSCAN: Voi epsilon toi uu, DBSCAN xac dinh duoc 2 cum chinh tuong tu K-Means, dong thoi nhan dang duoc cac diem noise (outlier) - day la cac listings co gia dac biet cao hoac cac thuoc tinh rat khac biet so voi phan con lai.

5. Danh Gia va So Sanh Hieu Qua Phan Cum

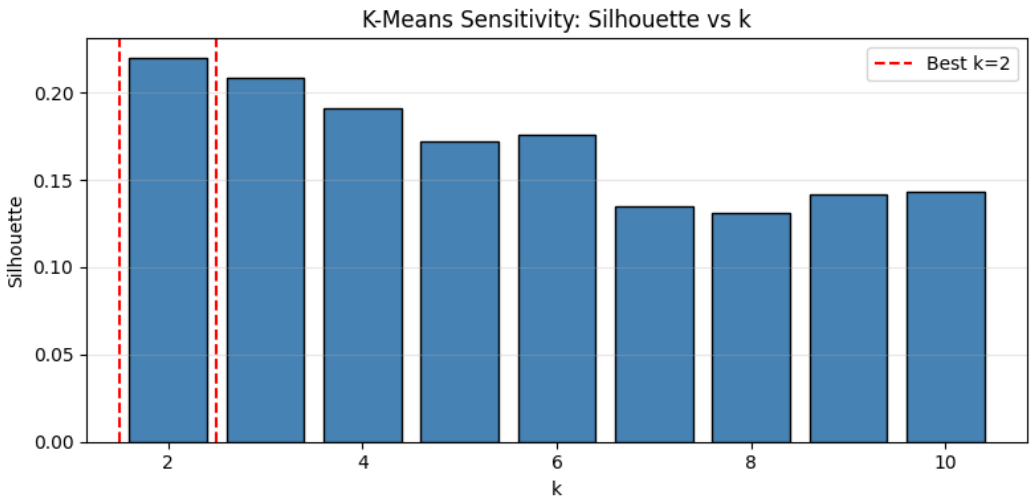
5.1 Bang So Sanh Cac Chi So Danh Gia

Chi so	K-Means	DBSCAN	Tot hon
Silhouette Score (cao tot hon)	0.2243	0.6082	DBSCAN
Davies-Bouldin Index (thap tot hon)	1.6809	0.3427	DBSCAN
Calinski-Harabasz Index (cao tot hon)	2560.94	77.29	K-Means
So cum xac dinh	2	2	-
Noise points	0	25	DBSCAN (detect outlier)



Hinh 12: So sanh truc quan cac chi so danh gia giua K-Means va DBSCAN

5.2 Phan Tich Do Nhay Tham So K-Means



Hinh 13: Silhouette Score cua K-Means theo cac gia tri k khac nhau

6. Thao Luan va Ket Luan

6.1 Insight Tu Ket Qua Phan Cum

Ca hai thuat toan deu nhat quan xac dinh **2 nhom chinh** trong du lieu NYC Airbnb:

Nhom 1 - Individual Hosts (Cluster 0): Phan lon listings (86%) thuoc ve cac chu nha ca nhan co 1-3 properties. Gia trung binh ~$194/dem, tu le reviews cao, availability trung binh. Day la phan khuc thi truong chinh cua Airbnb.

Nhom 2 - Professional Property Managers (Cluster 1): Nhom nho hon (14%) gom cac host chuyen nghiep quan ly hang chuc den hang tram listings. Gia co the thap hon nhung ho co quy mo lon va hoat dong co tinh kinh doanh ro rang.

6.2 Uu - Nhuoc Diem Tung Thuat Toan

	K-Means	DBSCAN
UU DIEM	+ Don gian, de hieu + Hieu qua tren du lieu lon + Ket qua on dinh, tai hien duoc	+ Tu dong chon so cum + Phat hien outlier hieu qua + Xu ly cum phi tuyen
NHUOC DIEM	- Phai chon k truoc - Nhay cam voi outlier - Chi phu hop cum hinh cau	- Kho chon epsilon, min_samples - Kem hieu qua voi du lieu cao chieu - Cham tren du lieu lon

6.3 Ket Luan

Qua qua trinh phan cum va phan tich du lieu NYC Airbnb voi 20,758 mau va 28 features, co the rut ra cac ket luan sau:

(1) **K-Means** la thuat toan phu hop hon cho bo du lieu nay vi cau truc du lieu co dang cum kha ro rang, phan phoi kha deu, va quy mo lon yeu cau hieu suat tinh toan tot. Silhouette Score = 0.2243 cho thay cac cum duoc tach biet o muc vua phai.

(2) **DBSCAN** tuy dat Silhouette Score cao hon (0.608 tren mau nho) nhung rat nhay cam voi tham so va kho ap dung tren toan bo du lieu lon (20k mau x 28 chieu). Uu diem chinh la kha nang phat hien outlier.

(3) **Insight chinh**: Thi truong Airbnb NYC co su phan chia ro rang giua Individual Hosts (ca nhan) va Professional Property Managers (doanh nghiep). Phan tich nay co the giup Airbnb dieu chinh chinh sach phi va quy dinh phu hop.

Khuyen nghi: Nen su dung K-Means (k=2) lam thuat toan phan cum chinh, ket hop DBSCAN de phat hien outlier. Co the thu nghiem them voi GMM (Gaussian Mixture Models) de co phan tich xac suat hon.